

Academic Programs

PEOs, POs & PSOs

B.Tech EEE

Program Educational Objectives (PEOs)

The student in Electrical and Electronics Engineering will
be able to:

Excel in chosen career and/or higher education with
technical competency in Electrical and Electronics
Engineering and allied engineering disciplines.

Demonstrate multidisciplinary skills and professional ethics
in providing sustainable solutions for engineering issues
through innovative product design and services to broader
societal context.

Work effectively as an individual, team member and/or
entrepreneur with good managerial and Communication skills.

Engage in lifelong learning to maintain and enhance
professional skills to align with changing societal needs.

Program Outcomes (POs)

Engineering Graduates will be able to:

Engineering knowledge: Apply the knowledge
of mathematics, science, engineering fundamentals, and an
engineering specialization to the solution of complex
engineering problems.

Problem analysis: Identify, formulate,
review research literature, and analyze complex engineering
problems reaching substantiated conclusions using first
principles of mathematics, natural sciences, and engineering
sciences.

Design/development of solutions: Design
solutions for complex engineering problems and Design System
components or processes that meet the specified needs with
appropriate consideration for the public health and safety,
and the cultural, societal, and environmental
considerations.

Conduct investigations of complex problems:
Use research-based knowledge and research methods including
design of experiments, analysis and interpretation of data,
and synthesis of the information to provide valid
conclusions.

Modern tool usage: Create, select, and
apply appropriate techniques, resources, and modern
engineering and IT tools including prediction and modelling

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to complex engineering activities with an understanding of the limitations.

The engineer and society: Apply reasoning

informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

Environment and sustainability: Understand

the impact of the professional engineering Solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

Ethics: Apply ethical principles and commit

to professional ethics and responsibilities and norms of the engineering practice.

Individual and team work: Function

effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

Communication: Communicate effectively on

complex engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write Effective reports and design documentation, make effective presentations, and give and receive clear instructions.

Project management and finance: Demonstrate

knowledge and understanding of the engineering and management principles and Apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

Life-long learning: Recognize the need for,

and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

Apply mathematics, basic sciences and electrical engineering

fundamentals to solve technical problems with the background of multi-disciplinary knowledge.

Identify, formulate, research literature and analyze complex

electrical and electronics Engineering problems attaining reasonable conclusions using fundamentals of mathematics, basic and engineering sciences.

Design solutions for complex electrical and electronics

engineering problems and the process to attain the specified solutions with societal, environmental and safety

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considerations

Create, select and apply modern tools to carry out complex electrical and electronics engineering activities with an understanding of the limitations

PEOs, POs

M.Tech (Power Electronics)

Proficient in applying sustainable and inclusive technologies to analyse, formulate and provide solutions for real time problems in diversified fields

Solve complex technological problems using emerging technologies and tools

Work effectively as an individual and team member with good communication skills in project execution

Demonstrate interdisciplinary skills and professional ethics in relating engineering issues to broader societal context

Engage in life long learning for a successful professional career.

An ability to independently carry out research /investigation and development work to solve practical problems in Power Electronics and Allied Fields.

An ability to write and present a substantial technical report/document.

An ability to demonstrate a degree of mastery over the Power Electronics and Systems.

Design and conduct experiments, as well as analyze the power electronic converters & drives and interpret the data.

Function on multidisciplinary technological issues related to power electronics advancements.

Use the techniques, skills, and modern engineering simulation tools necessary for the design and development of power converter topologies and engage in life long learning.

M. Tech (Power Systems)

PEO-1: Proficient in applying sustainable and comprehensive technologies to analyse, formulate and provide solutions for realtime problems in diversified fields.

PEO-2: Solve complex technical problems using emerging technologies and tools.

PEO-3: Demonstrate interdisciplinary skills and build hands on models for testing and research.

PEO-4: Develop professional ethics, strong communication skills, knowledge of social impacts and leadership qualities.

PEO-5: Engage in lifelong learning for a successful professional career.

PO-1: Ability to independently carry out research/investigation and Development work to solve practical problems.

PO-2: Ability to write and present a substantial technical report/document.

PO-3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program.

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PO-4: Design and conduct experiments, to analyse the power system models and interpret the data.

PO-5: Function on multidisciplinary technological issues assimilating power system advancements.

PO-6: Use the techniques, skills, and modern engineering simulation tools necessary for the design and development of power system technologies and engage in lifelong learning.